

Nonexistence of consecutive powerful triplets around cubes with mixed prime factorizations

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Abstract

A positive integer is *powerful* if every prime in its factorization occurs with exponent at least two. The Erdős–Mollin–Walsh conjecture asserts that no three consecutive integers are all powerful. Chan (2025) excluded triples of the form $x^3 - 1 = p^3 y^2$, x^3 , $x^3 + 1 = q^3 z^2$, and She (2025) excluded $x^3 - 1 = p^2 a^3$, x^3 , $x^3 + 1 = q^2 b^3$, with p, q prime. We close the remaining *mixed* configurations: there are no consecutive powerful triples of the form $x^3 - 1 = p^2 a^3$, $x^3 + 1 = q^3 z^2$, nor of the form $x^3 - 1 = p^3 y^2$, $x^3 + 1 = q^2 b^3$. The case analysis reduces both families to the single Diophantine equation $t^6 + t^3 + 1 = 3w^2$, which we solve completely: exploiting the palindromic symmetry of the genus-2 curve $3w^2 = t^6 + t^3 + 1$, we descend to an elliptic quotient of Mordell–Weil rank 0 (curve 1296.k1 in the LMFDB, Cremona label 1296f2), so the only rational solutions are $(t, w) = (1, \pm 1)$. We further show that no finite congruence sieve can decide this equation, so that some global (non-congruence) argument is required. As a complementary result we refine Beckon’s mod-36 constraint on the smallest member of a hypothetical triple to arbitrary prime-square moduli and quantify the density decay of admissible residue classes.

1 Introduction

A positive integer m is *powerful* if $p^2 \mid m$ for every prime $p \mid m$; equivalently $m = a^2 b^3$ with $a, b \in \mathbb{Z}_{>0}$ [7]. Pairs of consecutive powerful numbers, such as $(8, 9)$ and $(288, 289)$, occur infinitely often (e.g. via the Pell equation $x^2 - 8y^2 = 1$). In contrast, Erdős [5] conjectured, and Mollin and Walsh [9, 10] investigated systematically, that *no three consecutive integers are all powerful*. The conjecture remains open; under the *abc*-conjecture at most finitely many triples exist.

If $(n, n + 1, n + 2)$ is such a triple, the middle member may or may not be a perfect cube. The case in which the middle member is a cube x^3 has attracted recent attention. Chan [4] proved that there is no triple of the shape

$$x^3 - 1 = p^3 y^2, \quad x^3, \quad x^3 + 1 = q^3 z^2 \quad (p, q \text{ prime}),$$

and She [12] proved the analogous statement for

$$x^3 - 1 = p^2 a^3, \quad x^3, \quad x^3 + 1 = q^2 b^3 \quad (p, q \text{ prime}).$$

Both works treat *symmetric* configurations: the two outer members carry the same shape ($p^3 \cdot \square$ on both sides in [4]; $p^2 \cdot \text{cube}$ on both sides in [12]). Further recent activity in this circle includes van Doorn's study of three-term arithmetic progressions of consecutive powerful numbers [15]. The natural *mixed* configurations have, to our knowledge, not been treated. We close them.

Theorem 1. *There are no integers $x \geq 2$, primes p, q , and integers a, b, y, z such that either*

$$(M1) \quad x^3 - 1 = p^2 a^3 \quad \text{and} \quad x^3 + 1 = q^3 z^2, \quad (1)$$

$$(M2) \quad x^3 - 1 = p^3 y^2 \quad \text{and} \quad x^3 + 1 = q^2 b^3. \quad (2)$$

In particular, no triple of consecutive powerful numbers $(x^3 - 1, x^3, x^3 + 1)$ has mixed outer factorizations of these shapes.

The proof is a case analysis in the spirit of [4, 12] (Sections 3–4): factoring $x^3 \mp 1 = (x \mp 1)(x^2 \pm x + 1)$ and tracking 3-adic valuations reduces all but one branch of each family to classical Mordell equations $Y^2 = X^3 + k$ with $k \in \{2, -2, 54, -54, -162\}$, whose integral points are completely known [6, 2, 8]. Both remaining branches lead to one and the same equation,

$$t^6 + t^3 + 1 = 3w^2, \quad (3)$$

with $t \geq 7$ in family (M1) and $t \leq -5$ in family (M2). Equation (3) concerns the ninth cyclotomic value $\Phi_9(t)$ and defines a curve of genus 2, outside the reach of the degree-4 reductions used in [4, 12]; moreover, we prove (Remark 14) that *no* finite congruence sieve can decide it. Instead we solve it completely:

Theorem 2. *The only solutions of $t^6 + t^3 + 1 = 3w^2$ in rational numbers are $(t, w) = (1, \pm 1)$.*

The proof (Section 5) exploits the palindromic symmetry $t \mapsto 1/t$ of the genus-2 curve $3w^2 = t^6 + t^3 + 1$: the curve admits two independent elliptic quotients over $\mathbb{Q}$, and one of them has Mordell–Weil rank 0 with trivial torsion — it is the curve 1296.k1 in the LMFDB [8], Cremona label 1296f2 — which pins down all rational points. Since $t = 1$ lies in neither branch range, both remaining branches are empty and Theorem 1 follows.

As a complementary result in the same circle of ideas we record a modular refinement of a constraint of Beckon [1] on arbitrary consecutive powerful triples (not assuming a cube in the middle).

Proposition 3. *Let $(n, n + 1, n + 2)$ be powerful. Then, beyond Beckon's constraint $n \bmod 36 \in \{7, 27, 35\}$:*

- (i) *$n \bmod 900$ lies in an explicit set of 39 residue classes, and $n \bmod 44100$ in an explicit set of 1209 classes;*
- (ii) *for every prime $\ell \geq 5$, the proportion of admissible residues modulo ℓ^2 is $(\ell^2 - 3\ell + 3)/\ell^2$, while it is $1/4$ for $\ell = 2$ and $1/3$ for $\ell = 3$;*
- (iii) *consequently the density of admissible classes modulo $\prod_{\ell \leq L} \ell^2$ tends to 0 as $L \rightarrow \infty$, but only at the rate $\asymp (\log L)^{-3}$; in particular no finite sieve of this type can resolve the Erdős–Mollin–Walsh conjecture.*

All computer calculations below are short and reproducible; the elliptic-curve computations were performed in SageMath and independently cross-checked (two independent runs, plus the LMFDB data [8]); the symbolic identities were verified exactly in SymPy. The verification scripts are provided as ancillary files.

2 Preliminaries

Throughout, $x \geq 2$ is an integer, p, q are primes, and v_ℓ denotes the ℓ -adic valuation. We collect standard facts; proofs are short and included for completeness where not cited.

Lemma 4 (parity). *In any triple of consecutive powerful numbers the even member is divisible by 4. If the middle member is x^3 , then x is even; in particular $x^3 \pm 1$ are odd and coprime, so $p \neq q$ in (1)–(2).*

Proof. Exactly one or two of three consecutive integers are even; a powerful even number is $\equiv 0 \pmod{4}$. If n and $n+2$ were both even, one of them would be $\equiv 2 \pmod{4}$, hence not powerful. Thus the even member is unique, namely the middle one if the outer two are odd. For the middle member x^3 : if x were odd, then $x^3 \pm 1$ would be even, contradiction. Finally $\gcd(x^3 - 1, x^3 + 1) \mid 2$ and both are odd. $\square$

Lemma 5 (gcd and LTE). *$x^3 \mp 1 = (x \mp 1)(x^2 \pm x + 1)$ with $g_\mp := \gcd(x \mp 1, x^2 \pm x + 1) = \gcd(x \mp 1, 3) \in \{1, 3\}$. Moreover, if $3 \mid x \mp 1$ then $v_3(x^2 \pm x + 1) = 1$ exactly.*

Proof. $x^2 \pm x + 1 = (x \mp 1)(x \pm 2) + 3$, giving the gcd claim (cf. [12, Lemma 4]). For the valuation: if $x = 1 + 3k$ then $x^2 + x + 1 = 3 + 9k + 9k^2 \equiv 3 \pmod{9}$; if $x = -1 + 3k$ then $x^2 - x + 1 = 3 - 9k + 9k^2 \equiv 3 \pmod{9}$. In both cases $v_3 = 1$ exactly. $\square$

Lemma 6 (square sandwich). *For $x \geq 2$: $x^2 < x^2 + x + 1 < (x+1)^2$ and $(x-1)^2 < x^2 - x + 1 < x^2$. Hence neither $x^2 + x + 1$ nor $x^2 - x + 1$ is a perfect square.*

Lemma 7 (Chan-side structure). *Suppose $x^3 + 1 = q^3 z^2$ and $g_+ = 1$. Then $x+1$ is a perfect square and $x^2 - x + 1 = q^{3+2k} \cdot \square$ for some $k \geq 0$. Analogously, if $x^3 - 1 = p^3 y^2$ and $g_- = 1$, then $x-1$ is a perfect square and $x^2 + x + 1 = p^{3+2k} \cdot \square$.*

Proof. With $g_+ = 1$ the factors $x+1$ and $x^2 - x + 1$ are coprime, so each is, up to the prime q , a perfect square, and q^3 (odd exponent ≥ 3) divides exactly one of them. If q^3 divided $x+1$ with $x^2 - x + 1$ a square, Lemma 6 is contradicted. The mirrored statement is identical. $\square$

Lemma 8 (She's splitting; [12, Lemma 1]). *If $RS = p^2 C^3$ with $\gcd(R, S) = 1$, then one of R, S is a perfect cube and the other is p^2 times a perfect cube.*

Lemma 9 ([12, Lemma 2]). *$u^2 + u + 1 = 3v^3$ has only the integer solutions $u \in \{-2, 1\}$, and $u^2 - u + 1 = 3v^3$ only $u \in \{-1, 2\}$.*

Lemma 10 (Tzanakis; [14], cf. [12, Lemma 3]). *$u^2 + u + 1 = v^3$ has only $u \in \{-19, -1, 0, 18\}$, and $u^2 - u + 1 = v^3$ only $u \in \{-18, 0, 1, 19\}$.*

Lemma 11 (Mordell curves). *The integral points of $Y^2 = X^3 + k$ are: for $k = 2$: $(-1, \pm 1)$; for $k = -2$: $(3, \pm 5)$; for $k = 54$: $(3, \pm 9)$; for $k = -54$: $(7, \pm 17)$; for $k = -162$: none.*

Proof. All five values satisfy $|k| \leq 10^4$ and are covered by the complete solution of Mordell's equation in this range [6], extended to $|k| \leq 10^7$ in [2]; see also the corresponding LMFDB data [8]. (The case $k = -2$ is classical.) We re-verified all five lists with `integral_points` in SageMath in two independent runs. $\square$

We will also use without further comment: squares mod 9 are $\{0, 1, 4, 7\}$; cubes mod 9 are $\{0, 1, 8\}$; cubes of integers $\equiv 1 \pmod{3}$ are $\equiv \{1, 10, 19\} \pmod{27}$.

Exponent bookkeeping. For the Chan shape $x^3 \mp 1 = P^3 Y^2$ one has $v_P \equiv 1 \pmod{2}$, $v_P \geq 3$, and $v_r \equiv 0 \pmod{2}$ for primes $r \neq P$; for the She shape $x^3 \mp 1 = P^2 A^3$ one has $v_P \equiv 2 \pmod{3}$ and $v_r \equiv 0 \pmod{3}$ for $r \neq P$. Since $g_{\mp} \in \{1, 3\}$, every prime $r \neq 3$ divides exactly one of the two factors $x \mp 1$, $x^2 \pm x + 1$.

3 Family (M1): $x^3 - 1 = p^2 a^3$, $x^3 + 1 = q^3 z^2$

By Lemma 4, x is even and $p \neq q$. We split by $x \pmod{3}$.

Case 1: $x \equiv 0 \pmod{3}$ (so $g_- = g_+ = 1$)

By Lemma 7, $x + 1 = v^2$. By Lemma 8 applied to $(x - 1)(x^2 + x + 1) = p^2 a^3$:

- (ii) $x - 1 = p^2 u^3$, $x^2 + x + 1 = t^3$: Lemma 10 forces $x \in \{-19, -1, 0, 18\}$; only $x = 18$ is even and ≥ 2 , but $x - 1 = 17$ is prime, not of the form $p^2 u^3$. *Closed.*
- (i) $x - 1 = u^3$, $x^2 + x + 1 = p^2 t^3$: combining with $x + 1 = v^2$ gives $v^2 - u^3 = 2$, a point on $Y^2 = X^3 + 2$; by Lemma 11 $u = -1$, $x = 0 < 2$. *Closed.*

Case 2: $x \equiv 1 \pmod{3}$ (so $g_- = 3$, $g_+ = 1$)

Here $x^2 - x + 1 \equiv 1 \pmod{3}$, so $q \neq 3$, and $x + 1 = v^2$ by Lemma 7; on the She side $v_3(x^2 + x + 1) = 1$ exactly (Lemma 5).

- $p = 3$: here $v_3(x^3 - 1) = 2 + 3v_3(a)$, while $v_3(x^3 + 1) = v_3(x - 1) + 1$ by Lemma 5; hence $v_3(x - 1) = 1 + 3v_3(a)$ and $v_3(x^2 + x + 1) = 1$. All prime exponents in $x^2 + x + 1$ other than that of 3 are $\equiv 0 \pmod{3}$, so $x^2 + x + 1 = 3t^3$ with $3 \nmid t$; Lemma 9 gives $x \in \{-2, 1\}$, both < 2 . *Closed.*
- $p \neq 3$, $p \mid x - 1$: then again $x^2 + x + 1 = 3t^3$ ($v_3 = 1$, $p \nmid$, all other exponents $\equiv 0 \pmod{3}$), so $x \in \{-2, 1\}$. *Closed.*
- $p \neq 3$, $p \mid x^2 + x + 1$: then $v_3(x - 1) = 3v_3(a) - 1 \equiv 2 \pmod{3}$ and all other prime exponents in $x - 1$ are $\equiv 0 \pmod{3}$, hence $x - 1 = 9d^3$. But $x + 1 = v^2$ gives $v^2 = 9d^3 + 2 \equiv 2 \pmod{9}$, impossible since squares mod 9 are $\{0, 1, 4, 7\}$. *Closed.*

Case 3: $x \equiv 2 \pmod{3}$ (so $g_- = 1, g_+ = 3$)

She side (Lemma 8):

- (ii) $x - 1 = p^2 u^3, x^2 + x + 1 = t^3$: Lemma 10 gives $x \in \{-19, -1, 0, 18\}$; none is $\equiv 2 \pmod{3}$, even, and ≥ 2 . *Closed.*
- (i) $x - 1 = u^3$ with u odd, $u \equiv 1 \pmod{3}$ (so $u \equiv 1 \pmod{6}$), and $x^2 + x + 1 = p^2 t^3$. On the Chan side $g_+ = 3$ and $v_3(x^2 - x + 1) = 1$ exactly. Sub-split on q :
 - $q = 3$: $x^3 + 1 = 27z^2$ gives $v_3(x + 1) = 2 + 2v_3(z)$ even and all other exponents in $x + 1$ even, so $x + 1 = m^2$; then $m^2 - u^3 = 2$, again $Y^2 = X^3 + 2$, so $u = -1, x = 0 < 2$. *Closed.*
 - $q \neq 3, q^3 \mid x^2 - x + 1$: then $v_3(x + 1)$ is odd, say $x + 1 = 3^{2j+1}m^2$ with $\gcd(m, 3) = 1$ up to squares. If $j \geq 1$ then $27 \mid x + 1 = u^3 + 2$, i.e. $u^3 \equiv -2 \equiv 25 \pmod{27}$, impossible for $u \equiv 1 \pmod{3}$. Hence $j = 0, x + 1 = 3m^2$, i.e. $u^3 = 3m^2 - 2$; multiplying by 27: $(9m)^2 = (3u)^3 + 54$. By Lemma 11 the only points have $X = 3u = 3$, so $u = 1, x = 2$; but $x^3 - 1 = 7$ is not powerful. *Closed.*
 - $q \neq 3, q^3 \nmid x + 1$: then $x^2 - x + 1 = 3w^2$ ($v_3 = 1$ exactly, $q \nmid$, all other exponents even). With $x = u^3 + 1$:

$$(u^3 + 1)^2 - (u^3 + 1) + 1 = u^6 + u^3 + 1 = 3w^2,$$

i.e. equation (3) with $t = u$. Here $u \geq 1$ since $x \geq 2$, and $u = 1$ gives $x = 2$, where $x^3 - 1 = 7$ is not of the form $p^2 a^3$; together with $u \equiv 1 \pmod{6}$ this forces $t = u \geq 7$. *This is the only surviving branch of (M1); it is empty by Theorem 2.*

4 Family (M2): $x^3 - 1 = p^3 y^2, x^3 + 1 = q^2 b^3$

Again x is even, $p \neq q$; split by $x \pmod{3}$.

Case 1: $x \equiv 0 \pmod{3}$ ($g_- = g_+ = 1$)

Chan side (Lemma 7): $x - 1 = u^2$ with u odd. She side (Lemma 8) applied to $(x + 1)(x^2 - x + 1) = q^2 b^3$:

- (b) $x + 1 = q^2 s^3, x^2 - x + 1 = t^3$: Lemma 10 gives $x \in \{-18, 0, 1, 19\}$; no admissible value. *Closed.*
- (a) $x + 1 = s^3, x^2 - x + 1 = q^2 t^3$: then $s^3 - u^2 = 2$, i.e. a point on $Y^2 = X^3 - 2$; by Lemma 11 $(s, u) = (3, \pm 5), x = 26 \equiv 2 \pmod{3}$, contradicting the case. *Closed.*

Case 2: $x \equiv 1 \pmod{3}$ ($g_- = 3, g_+ = 1$)

She side has $g_+ = 1$:

- (b) $x + 1 = q^2 s^3, x^2 - x + 1 = t^3$: as above, no admissible x . *Closed.*

- (a) $x+1 = s^3$ with s odd, $s \equiv 2 \pmod{3}$, hence $s \equiv 5 \pmod{6}$ and $s \geq 5$; and $x^2 - x + 1 = q^2 t^3$. On the Chan side $v_3(x^2 + x + 1) = 1$ exactly. Sub-split on p :
- $p = 3$: then $v_3(x-1)$ is even and all other exponents in $x-1$ are even, so $x-1 = u^2$; then $u^2 = s^3 - 2$, so $(s, u) = (3, \pm 5)$ by Lemma 11, giving $x = 26 \equiv 2 \pmod{3} \neq 1$. *Closed.*
 - $p \neq 3, p \mid x-1$: then $x^2 + x + 1 = 3w^2$ ($v_3 = 1, p \nmid$, rest even). With $x = s^3 - 1$:
$$(s^3 - 1)^2 + (s^3 - 1) + 1 = s^6 - s^3 + 1 = 3w^2,$$
i.e. equation (3) with $t = -s \leq -5$. *This is the only surviving branch of (M2); it is empty by Theorem 2.*
 - $p \neq 3, p \mid x^2 + x + 1$: then $v_3(x-1)$ is odd, $x-1 = 3^{2j+1}m^2$. If $j \geq 1$ then $27 \mid s^3 - 2$, so $s^3 \equiv 2 \pmod{9}$, impossible (cubes mod 9). Hence $j = 0$, $s^3 = 3m^2 + 2$; multiplying by 27: $(9m)^2 = (3s)^3 - 54$ with $3 \mid X = 3s$. By Lemma 11 the only integral points have $X = 7$, not divisible by 3. *Closed.*

Case 3: $x \equiv 2 \pmod{3}$ ($g_- = 1, g_+ = 3$)

Chan side: $x-1 = u^2$ (u odd), $x^2 + x + 1 = p^{3+2k} \cdot \square$. She side: $v_3(x^2 - x + 1) = 1$ exactly. We first dispose of the case $q = 3$:

- $q = 3$: here $x^3 + 1 = 9b^3$, so $v_3(x^3 + 1) = 2 + 3v_3(b)$, while $v_3(x^3 + 1) = v_3(x + 1) + 1$ by Lemma 5; hence $v_3(x + 1) = 1 + 3v_3(b)$ and $v_3(x^2 - x + 1) = 1$, with all other prime exponents in $x^2 - x + 1$ being $\equiv 0 \pmod{3}$. Thus $x^2 - x + 1 = 3s^3$, and Lemma 9 gives $x \in \{-1, 2\}$; $x = 2$ fails since $x^3 - 1 = 7$ is not of the form $p^3 y^2$. *Closed.*
- $q \neq 3$: from $x^3 + 1 = q^2 b^3$ and $v_3(x^2 - x + 1) = 1$ we get $v_3(x + 1) = 3v_3(b) - 1 \equiv 2 \pmod{3}$, in particular $9 \mid x + 1$. Sub-split:
 - $q \mid x + 1$: then $x^2 - x + 1 = 3t^3$ ($v_3 = 1, q \nmid$, all other exponents $\equiv 0 \pmod{3}$), so $x \in \{-1, 2\}$ by Lemma 9; as before. *Closed.*
 - $q \nmid x + 1$: then $x + 1 = 3^{3v_3(b)-1} \cdot s^3 = 9c^3$ for an integer c , and $x - 1 = u^2$ gives $u^2 + 2 = 9c^3$; multiplying by 81: $(9u)^2 = (9c)^3 - 162$. By Lemma 11, $Y^2 = X^3 - 162$ has no integral points at all. *Closed.*

This completes the case analysis: both families reduce to the branches recorded above, all of which are empty except the two leading to (3).

5 The equation $t^6 + t^3 + 1 = 3w^2$

Write $\Phi_9(t) = t^6 + t^3 + 1$ for the ninth cyclotomic polynomial, so (3) reads $\Phi_9(t) = 3w^2$. The smooth projective model of

$$\mathcal{C} : Y^2 = 3t^6 + 3t^3 + 3 \quad (Y = 3w)$$

is a curve of genus 2. Since the right-hand side is palindromic, $\mathcal{C}$ carries the involution $\sigma : (t, Y) \mapsto (1/t, Y/t^3)$ defined over $\mathbb{Q}$, and hence (besides the hyperelliptic involution ι) the second involution $\tau = \sigma\iota$. The quotients $\mathcal{C}/\sigma$ and $\mathcal{C}/\tau$ are elliptic, and $\text{Jac}(\mathcal{C})$ is isogenous over $\mathbb{Q}$ to their product. Concretely:

Lemma 12. Set $u = t + 1/t$, and

$$V = \frac{Y(t+1)}{t^2}, \quad W = \frac{Y(t-1)}{t^2}.$$

Then on $\mathcal{C}$ one has the exact identities

$$V^2 = 3(u+2)(u^3 - 3u + 1) =: q_+(u), \quad W^2 = 3(u-2)(u^3 - 3u + 1) =: q_-(u).$$

Proof. Multiply $Y^2 = 3(t^6 + t^3 + 1)$ by $(t \pm 1)^2/t^4$ and use

$$(t \pm 1)^2(t^6 + t^3 + 1) = t^4(u \pm 2)(u^3 - 3u + 1),$$

which is a polynomial identity in t after substituting $u = t + 1/t$ (verified exactly; it follows from $u^3 - 3u + 2 = (u-1)^2(u+2)$ and $u^3 - 3u - 2 = (u+1)^2(u-2)$ together with $t^3 + t^{-3} = u^3 - 3u$ and $(t \pm 1)^2/t = u \pm 2$). $\square$

Proof of Theorem 2. Let $(t, w) \in \mathbb{Q}^2$ satisfy (3) and put $Y = 3w$. Note $t \neq 0$ (else $3w^2 = 1$) and $u = t + 1/t \neq -1$ (else $t^2 + t + 1 = 0$, no rational root). By Lemma 12, (u, W) is a rational point on the genus-one quartic curve

$$\mathcal{D}_- : W^2 = q_-(u) = 3(u-2)(u^3 - 3u + 1).$$

The quartic q_- is squarefree (its cubic factor has discriminant 81 and does not vanish at $u = 2$), so $\mathcal{D}_-$ is smooth of genus one; its two points at infinity satisfy $W_\infty^2 = 3$ and are conjugate over $\mathbb{Q}(\sqrt{3})$, hence not rational. $\mathcal{D}_-$ has the rational point $(2, 0)$.

We claim $\mathcal{D}_-(\mathbb{Q}) = \{(2, 0)\}$. Substituting $u = 2 + 1/T$, $W = S/T^2$ transforms $\mathcal{D}_-$ birationally into

$$S^2 = 9T^3 + 27T^2 + 18T + 3,$$

and the further substitution $(x, y) = (9T + 9, 9S)$ gives the elliptic curve

$$E : y^2 = x^3 - 81x + 243,$$

of conductor 1296. This is the curve **1296.k1** in the LMFDB [8] (Cremona label **1296f2**): it has Mordell–Weil rank 0 and trivial torsion, so $E(\mathbb{Q}) = \{\mathcal{O}\}$ — there is no affine rational point. (We verified rank 0 and triviality of torsion independently in SageMath with `rank(proof=True)`.) Explicitly, the maps are inverted by $T = (x - 9)/9$, $S = y/9$, so that $u = 2 + 1/T = (2x - 9)/(x - 9)$ and $W = S/T^2 = 9y/(x - 9)^2$; under this correspondence the affine rational points of $\mathcal{D}_-$ with $u \neq 2$ are in bijection with the affine rational points of E with $x \neq 9$. No rational points are lost at the exceptional loci: the two points at infinity of $\mathcal{D}_-$ are irrational (as noted above, $W_\infty^2 = 3$), the point $(2, 0)$ corresponds to $\mathcal{O} \in E$, and the points of E with $x = 9$ satisfy $y^2 = 243$ and are likewise irrational. Since $E(\mathbb{Q}) = \{\mathcal{O}\}$ contains no affine point, $u = 2$ is the only possibility, and then $W^2 = q_-(2) = 0$.

Pulling back to $\mathcal{C}$: $u = t + 1/t = 2$ forces $(t - 1)^2 = 0$, so $t = 1$ and $Y^2 = 9$, $w = \pm 1$. $\square$

Remark 13. The companion quotient $\mathcal{D}_+ : V^2 = q_+(u)$ has Jacobian $y^2 = x^3 - 189x + 999$ of rank 1 (LMFDB **1296.b1**, Cremona **129612**); this is why the descent must go through $\mathcal{D}_-$.

Remark 14 (no congruence sieve suffices). Equation (3) cannot be resolved by congruences alone, in the following precise sense. Substituting $z = t^3$, any solution gives a point on the conic $z^2 + z + 1 = 3w^2$; writing $2z + 1 = 3c$, $2w = b$ reduces the conic to the Pell equation $b^2 - 3c^2 = 1$, whose solutions with b even form a single orbit $\{z_k\}_{k \in \mathbb{Z}}$ of the recurrence $z_{k+1} = 14z_k - z_{k-1} + 6$ through $(z_0, z_1) = (-2, 1)$, the sign choice corresponding to the symmetry $z \mapsto -1 - z$. Asking whether (3) has a solution with $t \neq 1$ is asking whether the orbit contains a cube other than $z_1 = 1$. Since $z_1 = 1$ is a cube, its residue class modulo any period of the sequence is a “shadow” that no congruence can eliminate — and via the symmetry it also protects a class in the negative branch. Hence no finite system of congruence conditions can prove the nonexistence of further cubes in the orbit, and some global (here: elliptic) input is unavoidable. A direct check of the recurrence (in exact integer arithmetic) confirms that no z_k with $k \leq 2000$ other than z_1 is a cube, the largest candidates having more than 2200 decimal digits. (This computation only illustrates the covering obstruction; it is not used in the proof of Theorem 2, which rests entirely on the rank-0 descent of this section.)

Remark 15 (relation to Nagell–Ljunggren and to powers in recurrences). Since $\Phi_9(t) = (t^9 - 1)/(t^3 - 1)$, equation (3) is adjacent to the classical equations $(x^n - 1)/(x - 1) = 3y^m$ of Nagell–Ljunggren type. Note however that Nagell’s theorem on $x^2 + x + 1 = 3y^n$ ($n > 2$) does not apply: in (3) the square sits on the right-hand side ($m = 2$), while the cube condition sits inside the conic variable $z = t^3$. In the language of Remark 14, (3) asks for the cubes in a fixed binary recurrence orbit; the effective methods underlying the finiteness results of Pethő [11] and Shorey–Stewart [13] guarantee that there are only finitely many such cubes, but do not by themselves produce the explicit solution set. For the closely related factorization $(2z + 1)(z^2 + z + 1) = y^\ell$ see Bennett, Patel, and Siksek [3]. We are not aware of a published complete determination of the integer (or rational) solutions of (3); the contribution here is the explicit resolution via the rank-0 elliptic quotient.

6 A modular refinement of Beckon’s condition

Beckon [1] observed that the smallest member n of a triple of consecutive powerful numbers must satisfy $n \equiv 7, 27, 35 \pmod{36}$; this elementary but useful constraint appears not to have been taken up in the recent literature. The underlying mechanism generalizes to any prime-square modulus.

Proof of Proposition 3. If m is powerful and ℓ prime, then $v_\ell(m) \neq 1$; in particular, if $m \equiv c \pmod{\ell^2}$ with $\ell \mid c$, $\ell^2 \nmid c$, then m is not powerful. Call $r \pmod{\ell^2}$ *admissible* if none of $r, r+1, r+2$ is $\equiv \ell c' \pmod{\ell^2}$ with $\ell \nmid c'$. For $\ell \geq 5$ the $\ell-1$ forbidden residues $\ell c'$ are pairwise at distance ≥ 5 , so the $3(\ell-1)$ shifted prohibitions are distinct, leaving $\ell^2 - 3(\ell-1) = \ell^2 - 3\ell + 3$ admissible classes. For $\ell = 2$: n must be odd and $n+1 \equiv 0 \pmod{4}$, leaving the class $3 \pmod{4}$ (1 of 4). For $\ell = 3$: the windows avoiding $\{3, 6\} \pmod{9}$ start at $r \in \{0, 7, 8\}$ (3 of 9). By CRT the admissible classes multiply: modulo 36 one obtains exactly Beckon’s $\{7, 27, 35\}$; modulo $900 = 2^2 3^2 5^2$ there are $3 \cdot 13 = 39$ classes; modulo $44100 = 2^2 3^2 5^2 7^2$ there are $39 \cdot 31 = 1209$. (The explicit lists were computed by two independent implementations with matching output.) Finally, $\prod_{5 \leq \ell \leq L} (1 - \frac{3}{\ell} + \frac{3}{\ell^2}) \asymp (\log L)^{-3}$ by Mertens’ theorem, which proves (iii). $\square$

7 Concluding remarks

Together with [4] and [12], Theorem 1 completes the four shape-combinations $\{p^3 \cdot \square, p^2 \cdot \text{cube}\}^2$ for consecutive powerful triples around a cube in which both outer members are divisible by a single controlling prime to an exact pattern. Natural next steps in this program include the relaxations $p = q$ and composite cofactors, and She’s question whether $x^n - 1$ can be powerful for $x > 1$, $n > 2$. The bielliptic reduction of Section 5 may be of independent use for other palindromic sextic obstructions in this circle of problems.

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